



**Tonia
EL BACHA**

CONTACT

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- Nice, France
- [Tonia El Bacha](#)
- [Tonia EB](#)

SKILLS

- Programming languages : Python, C, Java, Rust, Racket, Ocaml, R, scilab, matlab, JavaScript
- Database : SQL
- Reverse Engineering: Ghidra, IDA, Wireshark
- Tools : Docker, Git, Github API

SOFT SKILLS

- Public Speaking
- Problem Solving
- Time Management
- Teamwork
- Adaptability

LANGUAGES

- | | |
|---------|--------|
| French | Fluent |
| English | Fluent |
| Arabic | Native |
| Spanish | A2 |

CERTIFICATIONS

- Project Management (EURECOM)
- Handimangement (Amadeus)
- Lovable (Hackathon Edhec)

HOBBIES

- Dance and Yoga
- Travel and Pohotography

Seeking a Job in Security
Starting October 2026

PROFILE

Master's graduate in Digital Security at EURECOM, with a strong interest in cybersecurity and a solid background in Mathematics and Computer Science. Hardworking and determined, I believe that I create my own success and that it is up to me to continuously push myself further.

EXPERIENCE

Cybersecurity Intern

Feb2026-Aug2026

SAP Labs France - Mougins / France

- Developed an internal scanning tool utilizing a 4-stage pipeline (Scan, Persist, Classify, Assess) to track AI adoption across **20,000+** SAP GitHub repositories.
- Secured the software supply chain by creating data pipelines that automatically monitor third-party code.

AI Security Research Internship

Jun2025-Aug2025

EURECOM - Sophia Antipolis / France

- Developed a statistical post-processing framework that mathematically enforces Equalized Odds to guarantee demographic fairness in pre-trained models.
- Conducted research on Federated Learning, optimizing the complex trade-offs between differential privacy and fairness in decentralized systems.

Pre-professional Education Assistant

Sep2022- Aug2024

Raoul Dufy College - Nice / France

- Teaching mathematics to more than **200** middle school students, correcting exams and tutoring sessions - **576 hours**.

PUBLICATIONS

Privacy-preserving and Fairness-aware Secure Aggregation in Federated Learning

- T. El Bacha**, A. Akram, I. Harrouche, M. Önen. PSD 2026 (Main Conference), Cádiz, Spain

EDUCATION

Master in Digital Security

2024 - NOW

EURECOM - Sophia Antipolis / France

- Courses : Forensics, Mobile Security, Cryptography, Hardware and Wireless Security, Clouds, System Security, Machine Learning

Double Degree in Mathematics and Computer Science

2021 - 2024

Côte d'Azur University - Nice / France

- Top 3** in graduating class.

PROJECTS

Fairness for Privacy-preserving in Machine Learning

Integrating fairness and privacy in machine learning models (Semester Project) in Python.

MoodTrip (Personalized Travel App)

Building a web application that generates custom travel itineraries tailored to the user's mood and group dynamics with Lovable.

Compiler

Translating simplified C to three-address code; includes symbol table and basic optimizations in C.